Mathematical overview

Mirror equivalence: kinds keep their names. Rotation-only rings may form reflected pairs when flips are admitted. The six printed eight-rings form four rotation classes but two rotation-and-flip classes. Distinct three-mark cyclic gaps distinguish a gap cycle from its reversed cycle.

Neighbor constraints: with two kinds, a ring with different neighbors exists exactly at even lengths, by alternation. The closing pair obstructs odd lengths. With three fixed kinds, a five-ring has exactly six rotation classes of proper colorings: multiplicities 2,2,1, a choice of singleton kind and two alternating orientations of the other four spots.

Universal windows: on a face-up binary ring, each bead starts one clockwise k-letter window, wrapping around. All 2^k codes require at least 2^k beads. Four beads suffice for k=2, uniquely up to rotation; eight suffice for k=3 with exactly two rotation classes, which are mirror twins. These are de Bruijn cycles, not prime-length counting.

Children change an equivalence rule, enforce a neighbor rule and construct a shortest ring containing every local code. Grades 2-3 can match/build with supplied cards; Grades 4-5 can defend parity, shortest length and completeness. Named kinds are never permuted. Only P1 permits flipped equivalence; all later tasks keep rings face up.

Entry routes and readiness

Shared packet: Grades 2-5. Counter matching and accurate tracing copies support P1. P2-3 require retaining a closing-neighbor rule. P4-7 require clockwise reading of two or three consecutive beads including wraparound, and matching the supplied cards; arithmetic to eight suffices.

Optional K-1 entry is making alternating rings or matching a small mirror pair, after an adult demonstrates the action. Two-window work is appropriate only when the child independently reads consecutive pairs and crosses the closing gap; do not force three-window coverage.

Materials and preparation

Per pair: two flat ring mats, at least eight counters each of kinds A/B and five C, tracing paper/clear overlay, removable start/viewer, fixed reading-direction marker, pencils and folders for preserving patterns. Working spots are 24 mm across; the eight-ring adjacent centers are about 25 mm apart, intended for 20 mm counters. Compact reference/record rings are not working mats.

Supply four two-letter and eight three-letter code cards; start each candidate test with a fresh complete deck and remove one matching card at every clockwise start. An already removed card means a repeated window and immediate failure. Cuttable cards can be pooled; no child must copy a catalog. Six reference rings for P1 and small recording rings for P3/P7 preserve collections. Estimated preparation 15-20 minutes; viewer, overlay and actual counter fit have not been rehearsed. Five sets serve eleven children at three fixed tables, one adult per table; the upper trio rotates roles.

Launch together

Let children move counters. Keep an original ring still while a tracing copy turns and flips; point to one mismatch. Before local windows, enact the printed five-bead non-task example: from the marked B read clockwise B-A-B and match BAB; from the lower-left A, A-B-A crosses the closing gap. Remove one matching practice card; then reset the deck for a new test. The viewer moves; kind names and reading direction stay fixed.

First visit and return visits

P1, P2-3 or P4-5: a 20-35 minute first visit. P6-7: a 25-45 minute return after reliable window reading. These are estimates. Save a checked ring and its code check; the complete collection can wait.

Problem 1: Mirror twins and two matching rules

Entry, prerequisites and timing

Grades 2-3 build and compare the six references with an overlay; Grades 4-5 explain the gap distinction. Allow 20-30 minutes. A letter naming a reference is not a permanent marked bead.

Mathematical facts and solutions

P1, turns only: groups {a,c}, {b}, {d,f}, {e}. Turns and flips: {a,b,c}, {d,e,f}. Describing the reference clockwise from its top, using A=1 and B=0, the words are a=11010000, b=10000101, c=01101000, d=11001000, e=10001001, f=00110010. This indexing describes the printed reference, not a fixed home mark during matching.

The A-to-A cyclic gaps of a are 1,2,5; b has the reversed cyclic order. These three unequal gaps cannot be reversed by a cyclic shift, so they are different rotation classes but one flip class. The gaps for d/e are 1,3,4 and reversed. Gap multisets {1,2,5} and {1,3,4} differ, so the two larger groups cannot merge even by flipping. Reference c rotates a; f rotates d.

An overlay verifies a claimed match by aligning every spot and kind. A single mismatch refutes that particular motion, not every possible motion. A gap argument or complete motion test supplies the all-motions justification.

Hold-back hints

Track the three A counters first. Ask which gaps appear around them and what a flip changes about their order. Keep a saved original beside the testing copy.

Depth and further questions

Invent another mirror pair and check that the flip is not already a turn. With at most two marked spots a polygon reflection always survives, so a binary ring of length at most 5 has no mirror twins. This is a minority-set argument, not a new catalog requirement.

Facilitator notes

After P1 explicitly restore face-up, rotation-only rules. Color names never swap when comparing patterns.

Problem 2-3: Different neighbors, including the closing pair

Entry, prerequisites and timing

K-1 oral entry can attempt the two-kind rings; Grades 2-3 build the six three-kind classes; Grades 4-5 defend completeness. Allow 20-35 minutes or a separate visit. A partner checks adjacent pairs and the closing pair.

Mathematical facts and solutions

P2: the four- and six-rings work, by ABAB and ABABAB. The five-ring cannot. Once a first kind is chosen, each neighbor is forced to alternate. On odd length the final spot repeats the first kind, violating their closing adjacency; even length closes correctly. The two fixed-position alternations match by a turn.

P3: exactly six rotation classes, represented by ABABC, ABACB, ABCAC, ABCBC, ACACB, ACBCB. They are read clockwise from any chosen start and compared by rotations only. A third kind must actually appear, because the two-kind odd-ring obstruction remains.

No kind can occupy three of the five spots without adjacent repeats: on a cycle, each occurrence would require a separating different-kind spot, so three occurrences require at least three separators. Therefore multiplicities are 2,2,1. Choose the singleton kind in three ways and temporarily place it first. The remaining four spots must alternate the other two kinds, in two orientations. Thus 3 times 2 gives six. A unique singleton fixes the temporary start, so two different listed orientations cannot be rotations of each other.

There are 30 valid fixed-position words, five views for each of six rings. Reflections merge the two orientations with the same singleton kind, but P3 does not allow that merge. The general named-color count in fixed n-position rings is (c-1)^n+(-1)^n*(c-1), n>=3; it is adult context, not needed for this classification.

Hold-back hints

Make the closing gap visible with a removable viewer, without turning it into a permanent marked bead. For P3, ask how often one kind can occur without touching itself, then use the least frequent kind as a temporary start.

Depth and further questions

Keep a singleton kind fixed and compare the two orientations by a flip; this connects P3 back to P1. If retaining the general coloring formula, count paths from a fixed first kind: let S_j count j-letter paths ending in that kind and D_j those ending differently. Then S_(j+1)=D_j and D_(j+1)=(c-1)*S_j+(c-2)*D_j. With S_1=1,D_1=0, total proper n-rings is c*D_n, giving the stated formula.

Facilitator notes

Children may share candidate cases and pool a checkable six-ring collection. They need not each rebuild all six to engage in the classification.

Problem 4-7: Every short code as a local window

Entry, prerequisites and timing

Grades 2-3 start P4-5 with four cards, then P6 when wraparound reading is reliable. Grades 4-5 seek P7 completeness and the shortest bound. Allow 25-45 minutes, with three-window work as a return.

Mathematical facts and solutions

P4: AABB works; its four windows are AA,AB,BB,BA. Up to rotation this is the unique four-bead solution. Fix the unique AA window to start. To get both kinds and avoid another AA, the next bead must be B; to include BB the fourth is B. Its closing windows then give BA and AA.

P5: each bead starts one two-letter window. A ring shorter than four beads supplies fewer than four window starts and cannot contain four distinct cards. The printed four-ring mat is construction capacity, not itself a proof of shortest length.

P6: AAABABBB gives windows AAA,AAB,ABA,BAB,ABB,BBB,BBA,BAA, each once. A ring needs at least eight beads for eight different three-letter cards, so this witness is shortest.

P7: exactly two shortest rotation classes: AAABABBB and AAABBBAB. Flipping either ring gives a turn of the other, so the collection is closed under flips; allowing flips gives one class. To classify, start at the unique AAA window and at its first A. The next bead is B, otherwise AAA repeats. If the following bead is A, the forced completion is AAA B A B B B. If it is B, the forced completion is AAA B B B A B. In the first branch, AAA B A A would repeat AAB at the next B, or make a second AAA; following with B forces BB then BBB. In the second branch, another A before BBB would prevent its required window; BBB then forces A to avoid repeating BBB, and the final B avoids repeating AAA across the closing gap. Alternatively enumerate all 256 indexed binary eight-words and retain those with eight distinct windows; canonical rotations leave these two classes.

All windows read clockwise across the closing gap. Reversing direction can produce the other mirror class; it is not a match under P7's rotation-only convention. Duplicate windows are failure even if all cards appear elsewhere, because shortest eight starts must account for eight distinct cards.

Hold-back hints

For each candidate restore the complete code deck. Read once from every clockwise start and move its one matching card to the found pile; if it is already gone, the ring fails. Ask which window would repeat if a proposed next bead is chosen. Preserve one candidate and its final card check, not concurrent window tallies.

Depth and further questions

After discovery, a three-code abc can be viewed as an edge from pair ab to pair bc. Every code once is an Euler circuit in that directed graph. Introduce this only if it helps a new question, with a worked visual; it is not a prerequisite for the printed task. The lower bound 2^k applies to any binary window length, but existence and all-solution counts need their own arguments.

Facilitator notes

Avoid treating a removed start marker as an extra kind. Start anywhere for checking; the ring is unmarked and face up. Use the two small eight-ring records to preserve P7's complete two-class collection; the large mat can be reused.

Sources, checks and classroom record

Sources and adaptation

Current Week 33, Prime-length necklaces, student packets pp. 1-6 and adult guide were inspected. Base tasks concern unrestricted rotation classes, periods and prime-length divisibility; this encore uses mirror sorting, neighbor constraints and universal windows. The base adult six-bead mirror anecdote is not repeated as the new task.

The base Oregon Math 391 worksheet is cited in its guide for prime-length counting; its body was not newly read here. New finite classes, parity and window arguments are independently derived and checked rather than attributed to that worksheet.

Laura Givental, Maria Nemirovskaya and Ilya Zakharevich, Math Circle by the Bay, Preface printed pp. viii-x (local PDF pages 9-11): themes with deep context and multiple entry levels, manipulatives, returning to explanations, and no universal estimate for a theme's duration. These are reported principles. The encore tasks, pairing and readiness routes here are local proposals, not classroom results from that book.

Verification

The author recomputed the facts, solutions and extension arguments from the actual tasks and design kernels. Independent mathematical reviews are retained separately with the editable sources.

Mathematical, render and extracted-source rebuild results belong in the release-validation record. No physical fit or procedure pretest and no classroom pilot has occurred.

What to record after a visit

Record each investigation actually used, examples attempted, a saved construction or plan, and where each pair wants to resume. A theme can support several visits; count use by investigation, not by finishing pages.

Record the child's action or words first; write your interpretation separately. Note whether children retained the choices after the launch or needed an adult to operate the investigation. All readiness and timing estimates below are unpiloted hypotheses.

Observe whether a flip is accidentally used after P1, whether kinds are swapped, whether closing neighbors/windows are omitted, and whether supplied card removal lets the child own the construction.

Status

Draft bonus companion. Unpiloted. Digital and mathematical checks do not establish physical fit, procedural rehearsal, staffing success or classroom pacing. Physical pretests have not been performed. Base student pages and base facilitator guide are separate.